

Question number	Answer	Additional guidance	Mark
7(a)	An explanation that includes the following points: • non-polar solute can cross the membranes because lipid is non-polar (1) • polar molecules cannot cross the membrane because they {cannot interact with the non-polar lipid / cannot dissolve in the non-polar lipid / are repelled by the non-polar lipid} (1)	ACCEPT hydrophobic phospholipid / fatty acid tails soluble in the lipid ACCEPT hydrophilic / hydrophobic phospholipid / fatty acid tails insoluble in the lipid NB one is polar (cannot cross) and one is non-polar (can cross) = 1 mark if no correct explanation given	(2)

Question number	Answer	Additional guidance	Mark
7(b)	An explanation that includes two of the following points: • because phospholipids have non-polar {tails / fatty acids} (and polar heads) (1) • the tails will orientate themselves away from the aqueous environment on {each side / both sides} of the membrane (1) • a bilayer is (the only) {stable / appropriate} arrangement (1)	ACCEPT hydrophilic and hydrophobic ACCEPT water repelled by water	(2)

Question number	Answer	Additional guidance	Mark
7(c)	An answer that includes two of the following points: • saturated or unsaturated (phospholipids / fatty acids) (1) • number of {saturated / unsaturated} fatty acids (1) • number of {carbon carbon / CC} double bonds (1) • length of (fatty acids / hydrocarbon chains / tails) (1) • some are attached to {carbohydrates / proteins} (1)	ACCEPT some contain CC double bonds some do not / some are kinked and some are straight / have different ratios of C : H ACCEPT number of carbons (in tail) IGNORE different size / bigger side chain	(2)

Question number	Answer	Additional guidance	Mark
7(d)	An explanation that includes the following points: • channel proteins and protein {pumps / carriers} present (1) • proteins need to span the membrane to transport substances across it (1) • because channel proteins needed for (facilitated) diffusion (1) • protein {pumps / carriers} to actively transport molecules (across the membrane) (1)	ACCEPT carrier proteins	(2)

Question number	Answer	Additional guidance	Mark
7(e)	- actual thickness and measured thickness given in same units for thickness measurements of {5 / 5.5 / 6} mm - magnification given to two significant figures 220 000 / 2.2×10^5 240 000 / 2.4×10^5 260 000 / 2.6×10^5	22.8 nm and 5 000 000 / 5 500 000 / 6 000 000 nm 0.0000228 mm and 5 / 5.5 / 6 mm Bald answer of 220 000 / 2.2×10^5 / 240 000 / 2.4×10^5 / 260 000 / 2.6×10^5 = 2 marks of these three values given in wrong standard form to 2 sig figs = 1 mark of these three values to wrong order of magnitude but correctly presented = 1 mark	(2)

Question number	Answer	Additional guidance	Mark
7(f)	An explanation that includes the following points: (fluid) because the phospholipids (and proteins) can move (within the {membrane / bilayer}) (1) (mosaic) because the proteins are {scattered / embedded / randomly distributed} (throughout the {membrane / phospholipids}) (1)	IGNORE monolayer ACCEPT named proteins e.g. channel proteins IGNORE cholesterol	(2)